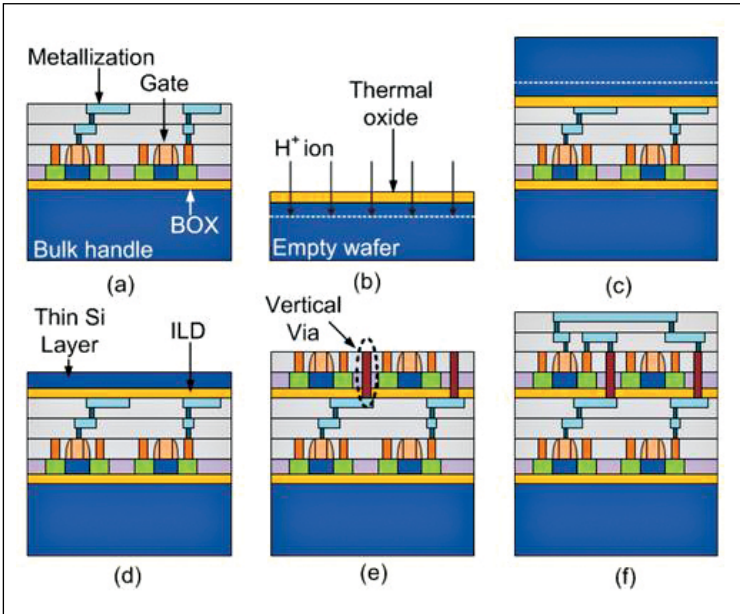




Just as the diagrams indicate, kinda technical.

This processing includes the creation of the interconnecting metal wires which are isolated by the dielectric layers. Traditionally, the insulating material has been a form of silicate glass. However, in recent times, the use of low dielectric constant materials have come to fore.

Wafer test

The demand for metrology in between different processing steps has gone up thanks to the heavily serialized nature of wafer processing. For instance, to have a strict control over the thickness of gate oxide, thin film metrology that's based on reflectometry or ellipsometry is used.

To verify that the wafers didn't get damaged by the processing steps that went before testing, they use the wafer test metrology equipment. If it's found that a large number of dies on a wafer have failed, the whole wafer gets scrapped- this way, the costs of further processing could be avoided. Virtual metrology-which uses statistical methods has been used to predict wafer properties. This negates having to perform any physical measurement.